Heatmaps & Clustered Displays

Workshop 5 — Module 4

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2025/26

Data Visualization for Data Scientists
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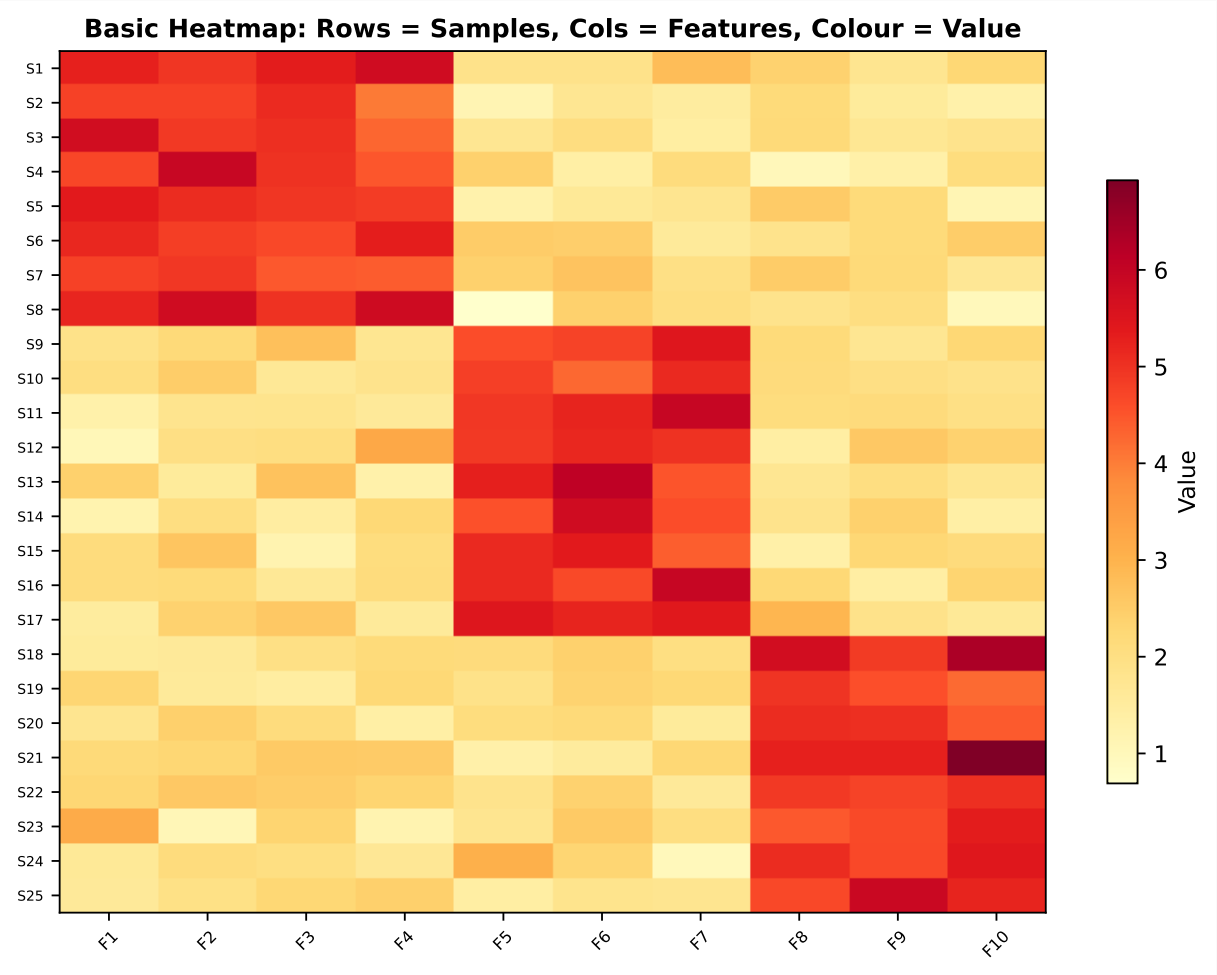
Module Roadmap

1. The heatmap as a multivariate display
2. Clustered heatmaps: dendrograms reorder rows and columns
3. Z-score normalisation: revealing within-row patterns
4. Five design principles for publication-quality heatmaps
5. Packages: pheatmap, ComplexHeatmap, sns.clustermap

Learning Outcome
You will produce clustered heatmaps with dendrograms and annotation tracks, apply z-score normalisation to reveal feature patterns, and choose between sequential and diverging palettes based on the data semantics.

The Heatmap

Rows × Columns, Colour = Value

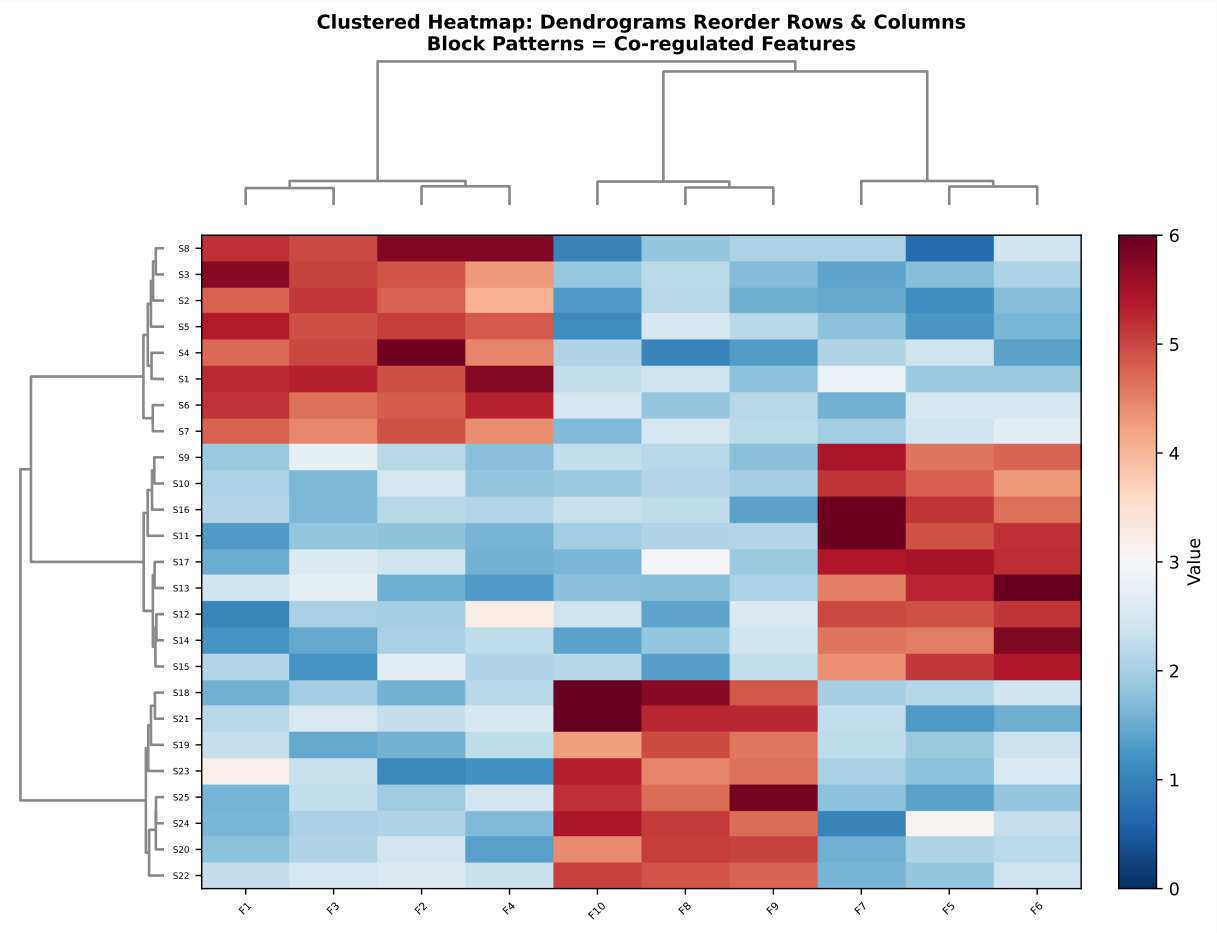


Data Insight

A heatmap encodes a matrix of values as coloured cells. Rows = observations (samples, genes, customers). Columns = variables (features, time points, metrics). Colour intensity = the value. Without clustering,

Clustered Heatmaps

Dendrograms Reveal Block Structure



Pro-Tip

Hierarchical clustering reorders both rows and columns so that similar profiles are adjacent. The resulting **block patterns** reveal which features co-vary and which samples share profiles. Without clustering, the same

Clustered Heatmap in Code

R (pheatmap)

```r
library(pheatmap)

pheatmap(mat,
  scale = "row",          # z-
      score
  clustering_method = "ward.D2",
  clustering_distance_rows =
    "euclidean",
  color = colorRampPalette(
    c("#1565C0","white","#E53935"
        ))(100),
  border_color = NA,
  fontsize_row = 6,
  fontsize_col = 8,
  annotation_row = ann_df, #
      metadata
  main = "Clustered Heatmap")

# ComplexHeatmap (Bioconductor)
library(ComplexHeatmap)
Heatmap(mat,
  name = "z-score",
  row_split = 3,       # k-means
      split
```

Python (seaborn)

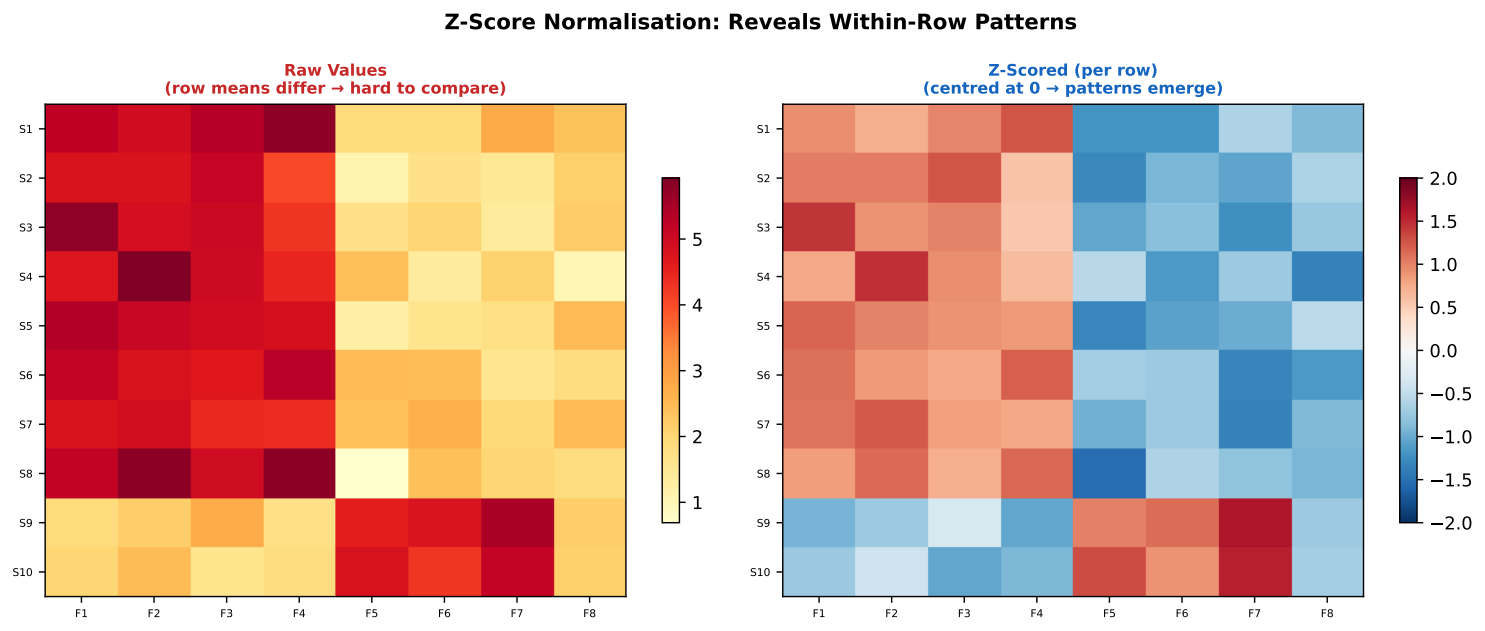
```python
import seaborn as sns

g = sns.clustermap(
    df,
    method="ward",
    metric="euclidean",
    standard_scale=0,     # z-
        score rows
    cmap="RdBu_r",
    figsize=(8, 10),
    linewidths=0.5,
    col_cluster=True,
    row_cluster=True,
    cbar_kws={"label": "z-score"
        },
    # Annotation sidebar
    row_colors=cluster_colors,
    col_colors=feature_colors)

g.ax_heatmap.set_xticklabels(
    g.ax_heatmap.get_xticklabels
        (),
    rotation=45, fontsize=7)
```

Z-Score Normalisation

Raw vs Z-Scored: Patterns Emerge



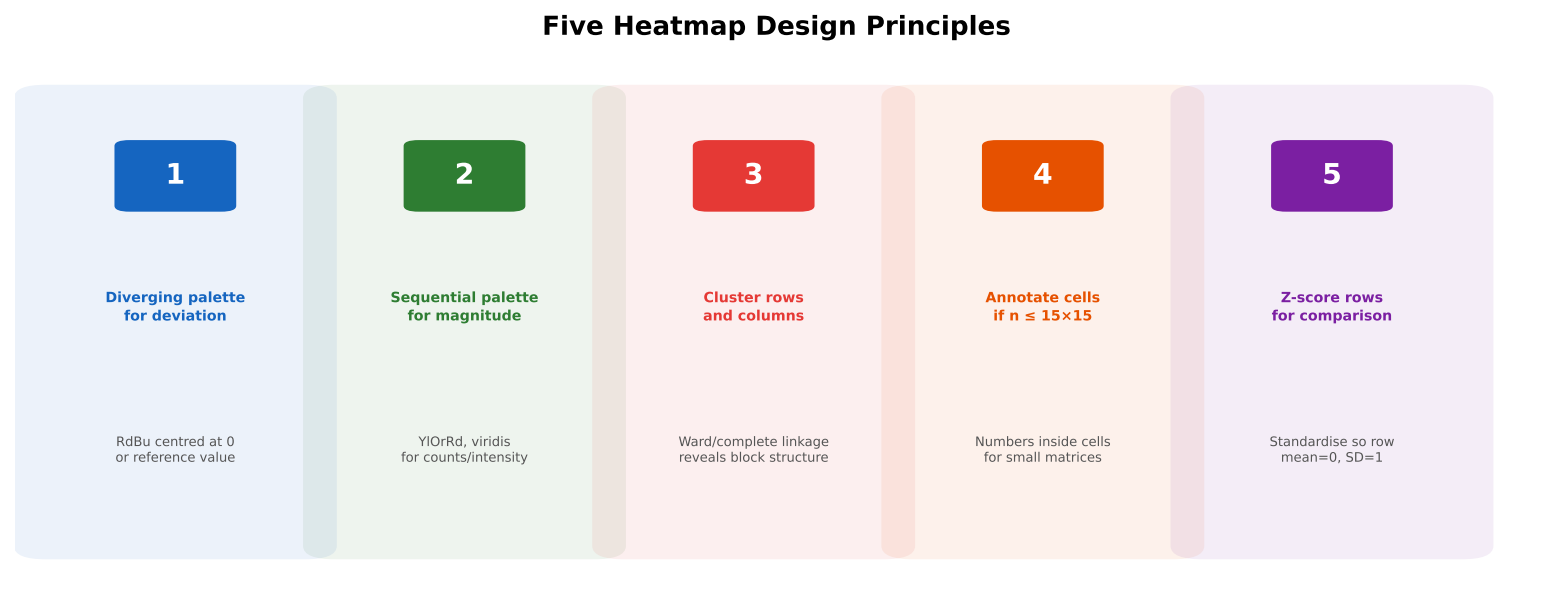
When to Z-Score

Situation	Normalise?	Palette
Variables on different scales	Yes (z-score rows)	Diverging (RdBu)
Absolute values matter	No	Sequential (YlOrRd)
Correlation matrix	Already $[-1, 1]$	Diverging (RdBu)
Count matrix (e.g., word freq)	Log-transform first	Sequential (viridis)
Comparing relative profiles	Yes (z-score rows)	Diverging

Z-scoring per row: $z_{ij} = (x_{ij} - \bar{x}_i)/s_i$. Centres each row at 0 with unit variance, so colour encodes "how many SDs above/below this row's mean" rather than the raw value.

Design Principles

Five Heatmap Design Principles



Packages

Heatmap Packages

Heatmap Packages: R & Python

Package	Lang	Call	Notes
`pheatmap`	R	`pheatmap(mat, cluster_rows=T)`	Most popular R heatmap; auto dendrograms
`ComplexHeatmap`	R	`Heatmap(mat) + HeatmapAnnotation()`	Bioconductor; multi-panel, annotation tracks
`sns.clustermap`	Python	`sns.clustermap(df, method='ward')`	seaborn; auto dendrograms + reordering
`plotly heatmap`	Both	`plot_ly(z=mat, type='heatmap')`	Interactive: hover for values

When to Use Heatmaps vs Other Charts

Data	Best Chart	Why Not
5–50 observations $\times$ 5–50 features	Heatmap	*This is the*
2 variables	Scatter	Heatmap w
1 variable over time	Line chart	Temporal a
$>$500 rows	Heatmap on summary stats	Too many
Correlation matrix	Heatmap (triangle)	*Ideal use ca*
Geographic data	Map	Spatial layo

Synthesis

Key Takeaways

1. Heatmaps encode a **matrix** as coloured cells. Their power comes from **clustering**: reorder rows and columns to reveal block patterns.
2. **Z-score rows** when variables have different scales; use **diverging palette** centred at 0.
3. **Sequential palette** (YlOrRd, viridis) for magnitude; **diverging** (RdBu) for deviation from a reference.
4. In R: `pheatmap` (quick), `ComplexHeatmap` (publication-quality, annotation tracks). In Python: `sns.clustermap` (auto dendrograms).
5. Sweet spot: 5–50 rows $\times$ 5–50 columns. Beyond that, aggregate or subset first.

Essential Readings

- Wilkinson, L. & Friendly, M. (2009). "The History of the Cluster Heat Map." *The American Statistician*, 63(2).
- Gu, Z. et al. (2016). "ComplexHeatmap." *Bioinformatics*, 32(18).
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapter 6.

Next Module: Introduction to Spatial Data (W05-M05)